

Network Security

Lecture 2a

How it all started



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Today!

How it all started

Why security is important?

Security concepts

Attacks on the different network layers

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Security?

- widespread use of data processing equipment:

computer security

- widespread use of computer networks and distributed computing systems:

network security



What is network security?

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What is network security?

- Access control
- Anti-malware
- Data loss prevention
- Email Security
- Web Security
- Network Security and the Cloud
- ...



First ever hack

- The Marconi Telegraph Prank
- 1903
- During a public demo of Marconi's new "secure" wireless telegraph, magician-turned-hacker Nevil Maskelyne decided to troll. He **interrupted the telegraph signal, sending the message "RATS" repeatedly**. The audience was stunned to say the least.

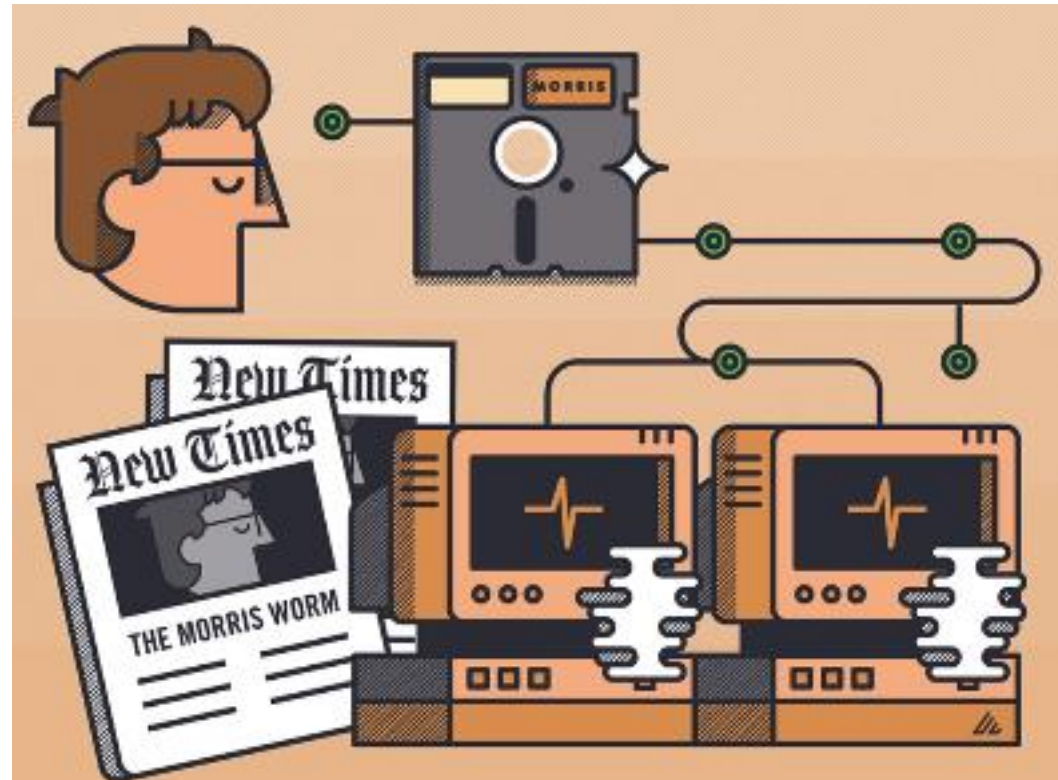


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Guglielmo Marconi



The day that changed everything





Brain (first computer virus for IBM)

Displacement	Hex codes	ASCII value
0000(0000)	FA E9 4A 01 34 12 00 07 14 00 01 00 00 00 00 20	-0J04↑●Π0
0016(0010)	20 20 20 20 20 20 57 65 6C 63 6F 6D 65 20 74 6F	Welcome to
0032(0020)	20 74 68 65 20 44 75 6E 67 65 6F 6E 20 20 20 20	the Dungeon
0048(0030)	20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20	
0064(0040)	20 20 20 20 20 20 20 20 20 20 20 20 20 20 20 20	
0080(0050)	20 28 63 29 20 31 39 38 36 20 42 61 73 69 74 20	(c) 1986 Basit
0096(0060)	26 20 41 6D 6A 61 64 20 28 70 76 74 29 20 4C 74	& Amjad (put) Lt
0112(0070)	64 2E 20 20 20 20 20 20 20 20 20 20 20 20 20 20	d.
0128(0080)	20 42 52 41 49 4E 20 43 4F 4D 50 55 54 45 52 20	BRAIN COMPUTER
0144(0090)	53 45 52 56 49 43 45 53 2E 2E 37 33 30 20 4E 49	SERVICES.. 730 NI
0160(00A8)	5A 41 4D 20 42 4C 4F 43 4B 20 41 4C 4C 41 4D 41	ZAM BLOCK ALLAMA
0176(00B0)	20 49 51 42 41 4C 20 54 4F 57 4E 20 20 20 20 20	. IQBAL TOWN
0192(00C0)	20 20 20 20 20 20 20 20 20 20 20 4C 41 48 4F 52	LAHDR
0208(0BD0)	45 2D 50 41 4B 49 53 54 41 4E 2E 2E 50 48 4F 4E	E-PAKISTAN..PHJN
0224(00E0)	45 20 3A 34 33 30 37 39 31 2C 34 34 33 32 34 3B	E :430791,443248
0240(00F0)	2C 32 38 30 35 33 30 2E 20 20 20 20 20 20 20 20	,280530.



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But why do we need security?



What was the largest DDoS attack of all time?

On June 1, a Google Cloud Armor customer was targeted with a series of HTTPS DDoS attacks which peaked at 46 million requests per second. This is the largest Layer 7 DDoS reported to date—at least 76% larger than the [previously reported record](#). To give a sense of the scale of the attack, that is like receiving all the [daily requests](#) to [Wikipedia](#) (one of the top 10 trafficked websites in the world) in just 10 seconds.

<https://cloud.google.com/blog/products/identity-security/how-google-cloud-blocked-largest-layer-7-ddos-attack-at-46-million-rps>

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- **2021 Yandex attack**
- **2021 Cloudflare attack**
- **February 2020 attack reported by AWS**
- **February 2018 GitHub DDoS attack**
- **2016 Dyn attack**
- **2013 Spamhaus attack**
- **2007 Estonia attack**
- **2000 Mafiaboy attack**

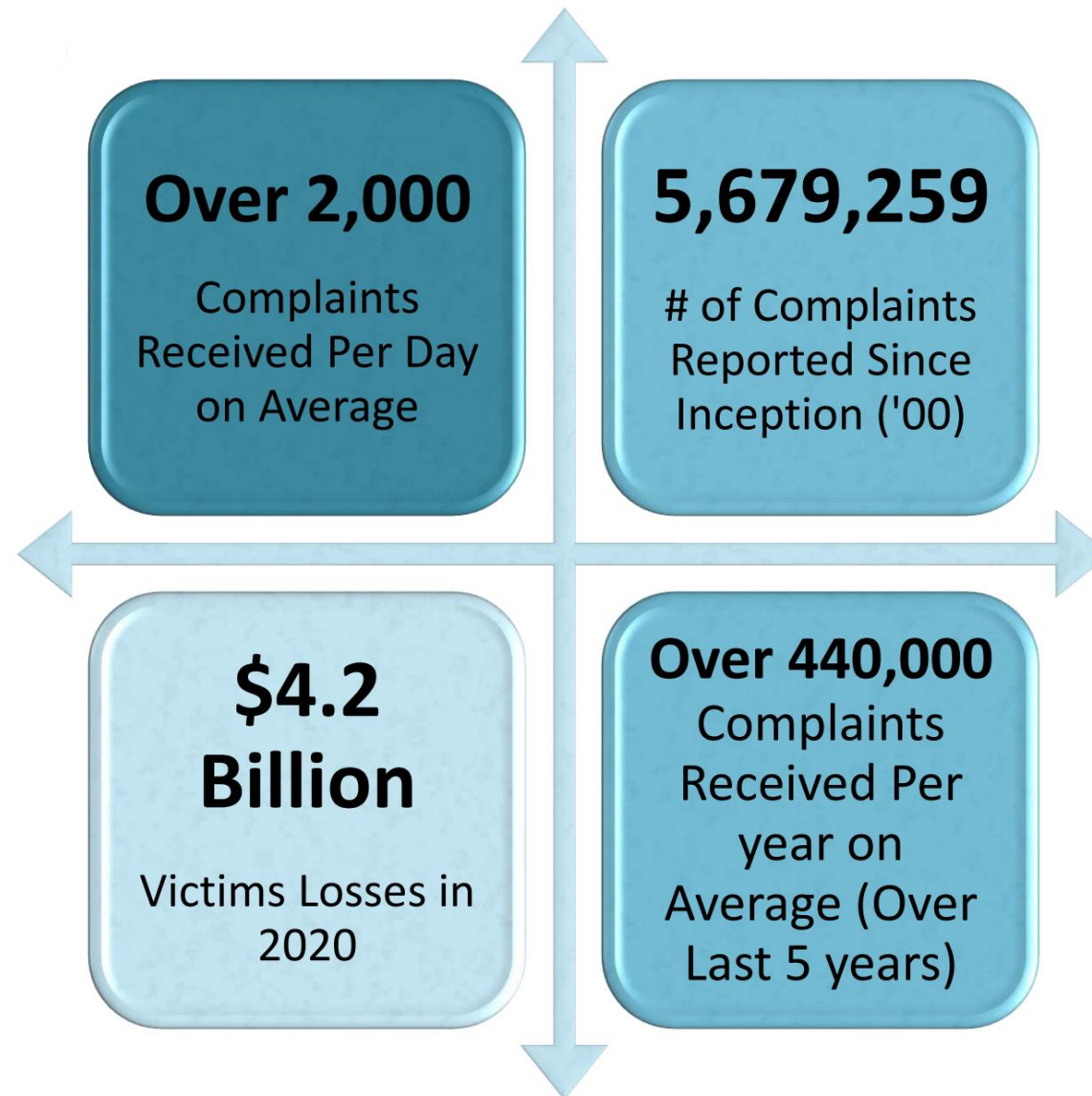
Jun 18, 2020 12:30:00

It turned out that AWS was under 2.3 Tbps DDoS attack



**IN CASE OF
CYBERATTACK

BREAK GLASS
AND PULL CABLES**



Sources: 2020 Internet Crime Report

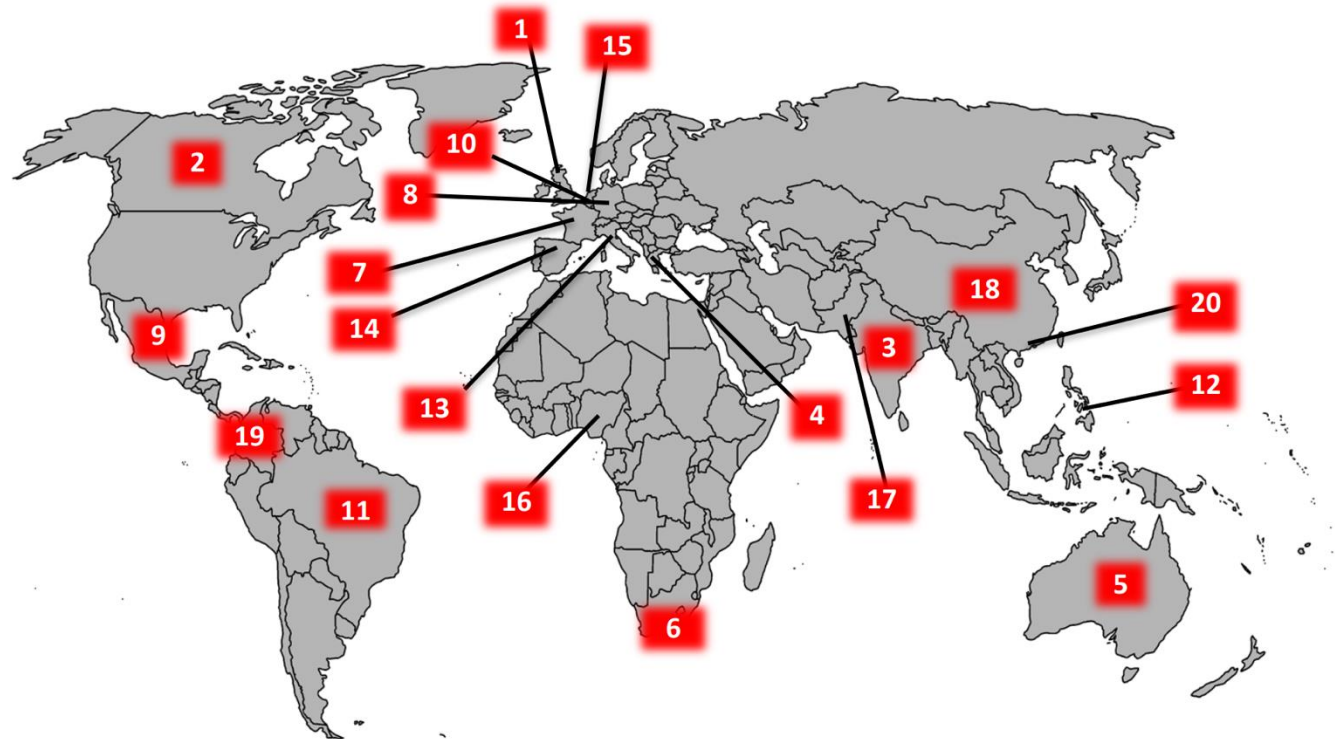
2020 VICTIMS BY AGE GROUP

Victims		
Age Range ⁷	Total Count	Total Loss
Under 20	23,186	\$70,980,763
20 - 29	70,791	\$197,402,240
30 - 39	88,364	\$492,176,845
40 - 49	91,568	\$717,161,726
50 - 59	85,967	\$847,948,101
Over 60	105,301	\$966,062,236

Source: 2020 Internet Crime Report

2020 - TOP 20 INTERNATIONAL VICTIM COUNTRIES

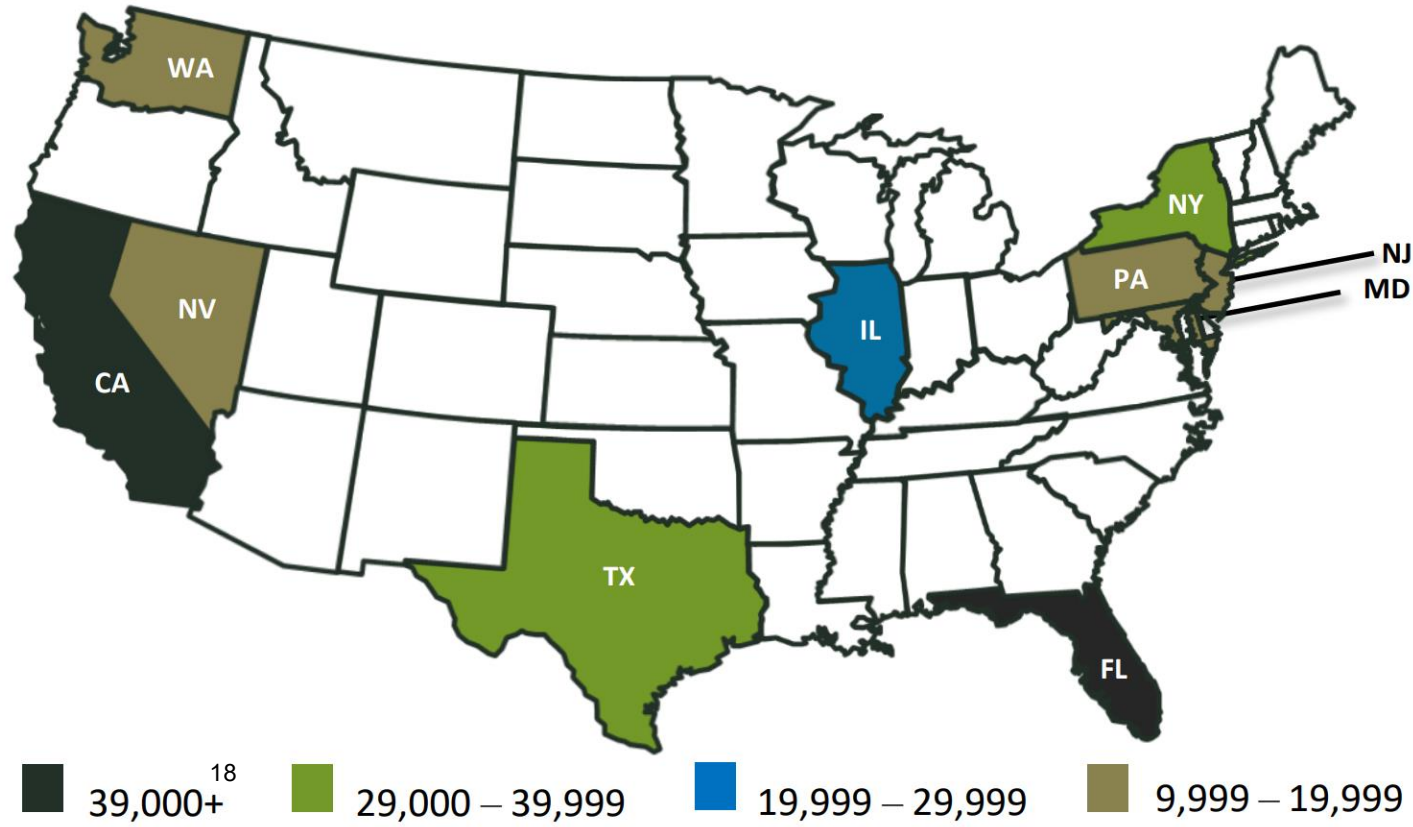
Excluding the United States⁸



1. United Kingdom	216,633	6. South Africa	1,754	11. Brazil	951	16. Nigeria	443
2. Canada	5,399	7. France	1,640	12. Philippines	898	17. Pakistan	443
3. India	2,930	8. Germany	1,578	13. Italy	728	18. China	442
4. Greece	2,314	9. Mexico	1,164	14. Spain	618	19. Colombia	418
5. Australia	1,807	10. Belgium	1,023	15. Netherlands	450	20. Hong Kong	407

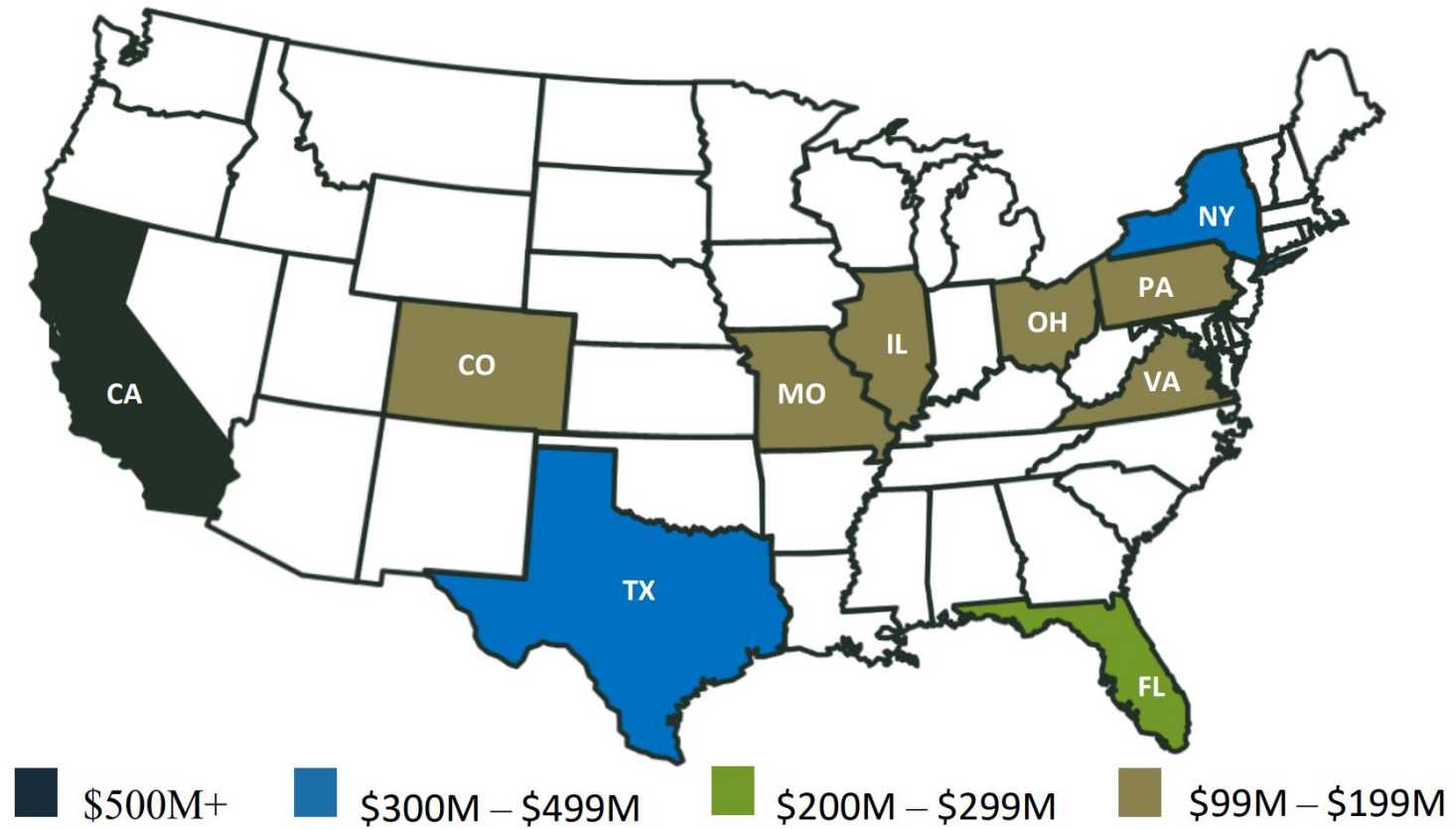
Source: 2020 Internet Crime Report

2020 - TOP 10 STATES BY NUMBER OF VICTIMS⁹



Source: 2020 Internet Crime Report

2020 - TOP 10 STATES BY VICTIM LOSS¹⁰



Sources: 2020 Internet Crime Report

2020 CRIME TYPES

By Victim Count			
Crime Type	Victims	Crime Type	Victims
Phishing/Vishing/Smishing/Pharming	241,342	Other	10,372
Non-Payment/Non-Delivery	108,869	Investment	8,788
Extortion	76,741	Lottery/Sweepstakes/Inheritance	8,501
Personal Data Breach	45,330	IPR/Copyright and Counterfeit	4,213
Identity Theft	43,330	Crimes Against Children	3,202
Spoofing	28,218	Corporate Data Breach	2,794
Misrepresentation	24,276	Ransomware	2,474
Confidence Fraud/Romance	23,751	Denial of Service/TDoS	2,018
Harassment/Threats of Violence	20,604	Malware/Scareware/Virus	1,423
BEC/EAC	19,369	Health Care Related	1,383
Credit Card Fraud	17,614	Civil Matter	968
Employment	16,879	Re-shipping	883
Tech Support	15,421	Charity	659
Real Estate/Rental	13,638	Gambling	391
Advanced Fee	13,020	Terrorism	65
Government Impersonation	12,827	Hacktivist	52
Overpayment	10,988		

Descriptors* 20		
Social Media	35,439	*These descriptors relate to the medium or tool used to facilitate the crime and are used by the IC3 for tracking purposes only. They are available as descriptors only after another crime type has been selected. Please see Appendix B for more information regarding IC3 data.
Virtual Currency	35,229	



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Source: 2020 Internet Crime Report

2020 Crime Types *Continued*

By Victim Loss

Crime Type	Loss	Crime Type	Loss
BEC/EAC	\$1,866,642,107	Overpayment	\$51,039,922
Confidence Fraud/Romance	\$600,249,821	Ransomware	**\$29,157,405
Investment	\$336,469,000	Health Care Related	\$29,042,515
Non-Payment/Non-Delivery	\$265,011,249	Civil Matter	\$24,915,958
Identity Theft	\$219,484,699	Misrepresentation	\$19,707,242
Spoofing	\$216,513,728	Malware/Scareware/Virus	\$6,904,054
Real Estate/Rental	\$213,196,082	Harassment/Threats Violence	\$6,547,449
Personal Data Breach	\$194,473,055	IPR/Copyright/Counterfeit	\$5,910,617
Tech Support	\$146,477,709	Charity	\$4,428,766
Credit Card Fraud	\$129,820,792	Gambling	\$3,961,508
Corporate Data Breach	\$128,916,648	Re-shipping	\$3,095,265
Government Impersonation	\$109,938,030	Crimes Against Children	\$660,044
Other	\$101,523,082	Denial of Service/TDoS	\$512,127
Advanced Fee	\$83,215,405	Hacktivist	\$50
Extortion	\$70,935,939	Terrorism	\$0
Employment	\$62,314,015		
Lottery/Sweepstakes/Inheritance	\$61,111,319		
Phishing/Vishing/Smishing/Pharming	\$54,241,075		

Descriptors*²¹

Social Media	\$155,323,073
Virtual Currency	\$246,212,432

*These descriptors relate to the medium or tool used to facilitate the crime and are used by the IC3 for tracking purposes only. They are available only after another crime type has been selected. Please see Appendix B for more information regarding IC3 data.

Source: 2020 Internet Crime Report

SolarWinds hack

- In early 2020, hackers secretly broke into Texas-based SolarWind's systems and added malicious code into the company's software system.
- The system, called "Orion," is widely used by companies to manage IT resources.
- SolarWinds has 33,000 customers that use Orion.



SolarWinds hack victims

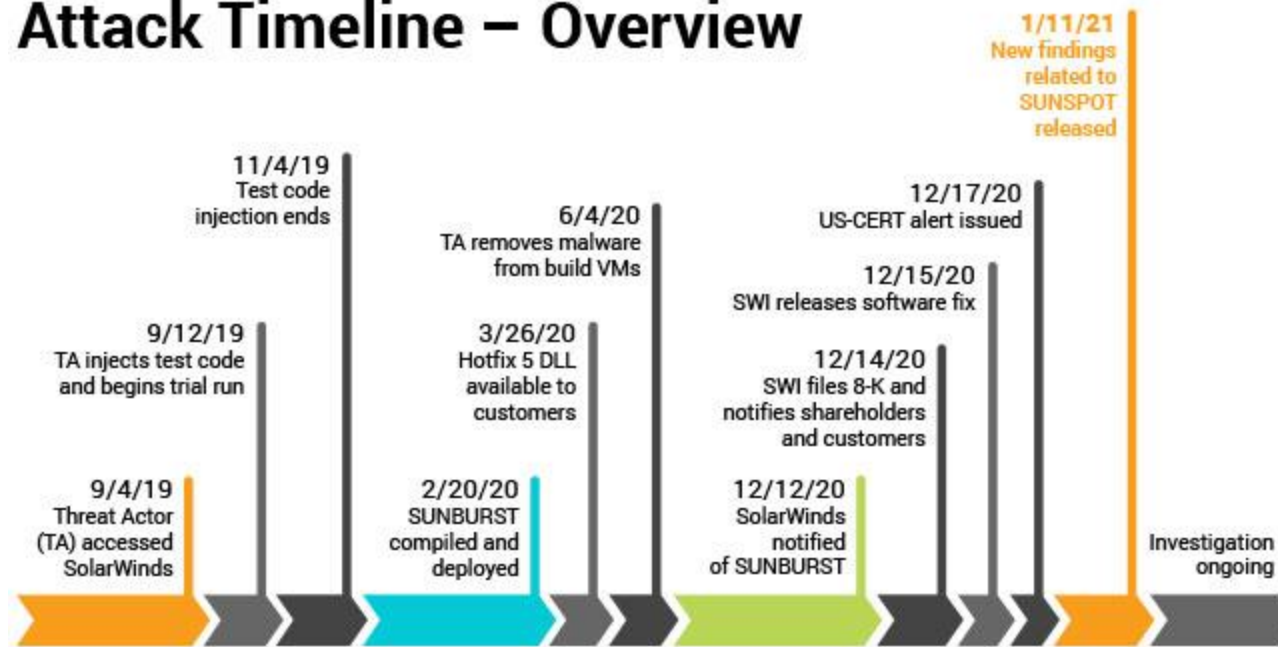


- US agencies
 - The Pentagon,
 - The Department of Homeland Security,
 - The State Department, the Department of Energy, the National Nuclear Security Administration, and the Treasury — were attacked.
- Private companies
 - Microsoft, Cisco, Intel, and Deloitte.
 - California Department of State Hospitals, and Kent State University.

SolarWinds hack hackers

- Russian Hackers backed by Russian government.
- Russia denies ties to the group.

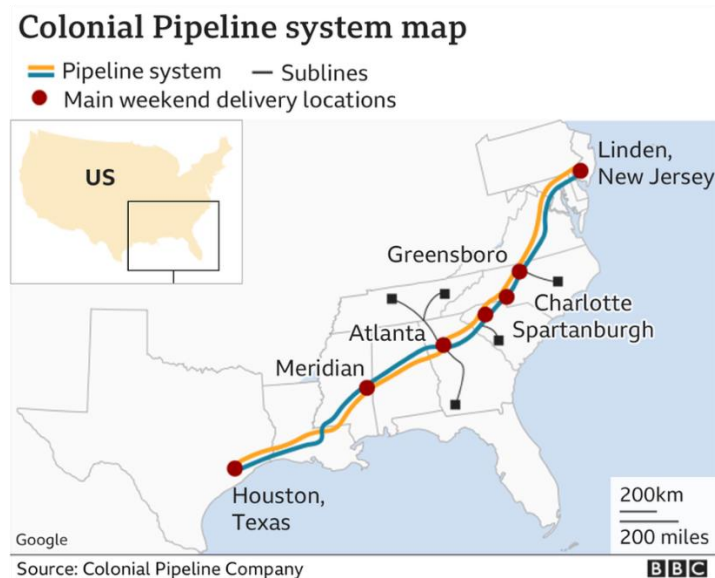
Attack Timeline – Overview



All events, dates, and times approximate and subject to change; pending completed investigation.

Colonial Pipeline cyber attack

- American oil pipeline system that originates in Houston, Texas, and carries gasoline and jet fuel mainly to the Southeastern United States.
- A ransomware cyberattack that impacted computerized equipment managing the pipeline.

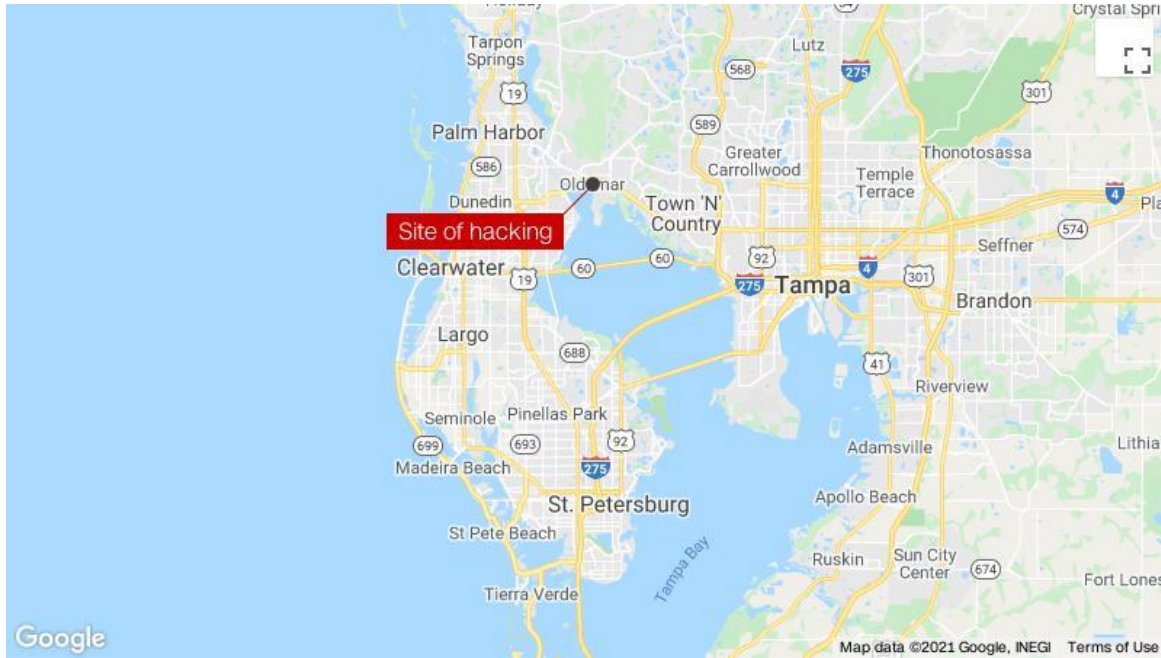


Colonial Pipeline cyber attack

- The FBI and various media sources identified the criminal hacking group DarkSide as the responsible party.
- 17 states and Washington, D.C. were affected.
- Colonial Pipeline paid the requested ransom (75 bitcoin or \$4.4 million) within several hours after the attack.



Water supply poisoning



- A computer hacker gained access to the water system of Oldsmar city.
- A computer controlling water treatment system was remotely accessed.
- A plant operator saw an attempt to access the system in the morning but assumed it was his supervisor.

Water supply poisoning

- The hacker accessed the treatment software and increased the sodium hydroxide content from 100 parts per million to 11
- The operator immediately reduced the level to normal.
- Swallowing sodium hydroxide at such high levels can cause damage to the mouth, throat and stomach and induce vomiting, nausea and diarrhea.



Düsseldorf hospital attack

- The attack encrypted about 30 hospital servers and left a message instructing the Heinrich Heine University, to which the Düsseldorf hospital is affiliated, to contact the attackers.
- A woman seeking emergency treatment for a life-threatening condition could not be offered treatment by the hospital.
- The woman was rushed to a hospital about 20 miles away, resulting in about a one-hour delay in treatment. **She died.**



Düsseldorf hospital attack

- Düsseldorf police communicated with the attackers and told them that the attack had hit a hospital treating emergency patients, not the university.
- The attackers provided a decryption key to unlock the servers.
- Hackers backed by the Chinese government



Example of an Attack



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What do you think?